



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-2026)
Faculty Name	Ms. G. Mamatha
Student Name	Brundavanam VR VR SrisaiHarini
Roll Number	2520030384
Branch	CSE
Course Outcome	CO-3

Case Study Topic: LibraryNet – Intelligent Digital Library & Knowledge Retrieval System

Work Done Summary:

In CO3, graph algorithms were implemented to manage relationships between books, citations, and topics. BFS and DFS were used for recommendation systems and citation analysis.

Implementation Details:

Graph Representation

Books are represented as nodes.

Citations and references are represented as edges.

Breadth First Search (BFS)

Used to explore related books.

Generates recommendations.

Depth First Search (DFS)

Detects cycles in citation networks.

Identifies repeated references.

Minimum Spanning Tree (MST)

Optimizes knowledge graph connections.

Removes unnecessary edgesThe system ensures efficient performance even when large volumes of patient data are added or removed frequently.

Result : Screenshots / Output

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Output
Python
AI
Data Science
Machine Learning
Statistics

=== Code Execution Successful ===
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Algorithm	Time Complexity
BFS Traversal	$O(V + E)$
DFS Traversal	$O(V + E)$
Minimum Spanning Tree	$O(E \log V)$

GitHub Link: <https://github.com/HariniBrundavanam/DSA-2-project>

Student Signature	Faculty Signature